

Computer Basics and PC Software

BCS-11

1 Question:-

(a) What are the functions of various operational units of a computer system? What is von Neuman Architecture? How can you relate von Neumann architecture to an actual computer? Explain with the help of an example configuration.

Answer:- A computer is an electronic device that accepts data, processes it and returns processed information (Result). It works on the principle of Input, process and output.

Input → Process → Output

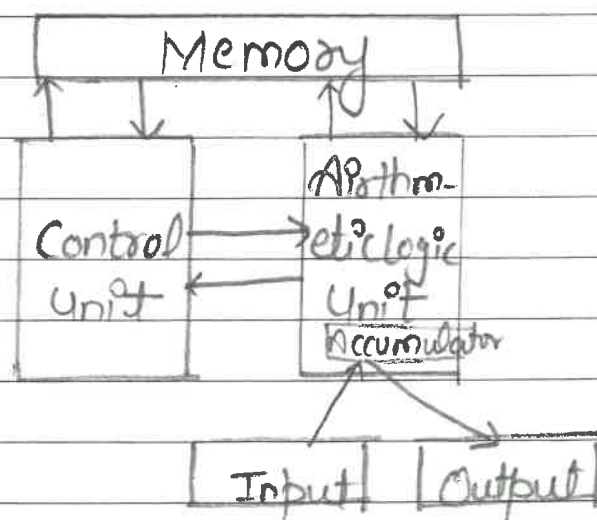
Main functional Unit:-

- CPU:- It is the brain of any computer system. It is just like brain that takes all major decisions, makes all sorts of calculations, functions and controlling the operations.
- The Arithmetic Logical Unit:- It is an important component of the CPU, which carry the actual execution of the Instructions.

- The Control unit :- It determines the sequence in which computer programs and instructions are executed. It acts like the supervisor seeing that things are done in proper fashion.
- A Von Neumann machine has single path between the main memory and control unit. This feature is known as bottleneck.

Structure of von-Neumann Computer machine & function

John Von Neumann architecture a design model for the modern computers, which has central processing unit (CPU), and the concept of memory used for storing both data and instructions. All computers have memory, an I/O system, arithmetic logic unit (ALU) and Control unit (CU).



(Structure of Von-Neumann Machine)

Features of von-Neumann

- CPU consists of ALU and CU
- Program and data are stored in the same memory unit.
- A von Neumann machine has unique memory address of each location.
- A von Neumann machine has only a single path between main memory and control unit this feature is called bottleneck.
- Serial processing system
- Input/output system
- Main memory system.

(b) Compare and contrast the characteristics and/or organization of the following:

(i) DRAM vs. SRAM

Answer:-

Static RAM (or SRAM)

- Static RAMs retain stored information for long time as the power supply is on.
- Static RAMs are costly and consume more power. Static RAMs have a higher speed than dynamic RAMs.
- Static Ram is faster and so is used in cache memory.
- SRAM does not need to be refreshed.

Dynamic RAM (or DRAM)

- Dynamic RAM loses its stored information in a very short time even the power supply is on.
- Dynamic RAMs are cheaper and consume less power.
- Dynamic RAM is cheaper and so is used for main memory.
- Dynamic RAM requires the data to be refreshed periodically.

(ii) Access time on Magnetic disks Vs. access time on Magnetic tapes.

Answer:-

Magnetic disks:- It provides random access. This means that a file can be accessed from any location directly. The total access time for a disk is equal to the seek time plus the latency time. The average access time for most disk systems is usually between 10 to 100 milliseconds.

Magnetic tapes:- A file or more particular information stored on a magnetic tape cannot be accessed directly on random basis as is possible in the case of hard-disks or floppy disks. These devices are slower, but due to their low cost, they are still widely used for massive data warehouse and other business storage requirements.

iii) Pen Drive vs. CD-RW

Answer: Pen drive is available as a very convenient and flexible data storage medium which can store up to 256 GB data. It can be used for the same purposes as floppy-disks or CD-ROMs. Pen Drives are a smaller, faster, durable and more reliable storage medium.

CD-R (compact-disk recordable) is another optical disk technology. The user can record their own data once on a CD with a CD-R disk drive Unit. After this recording user can read the data as many times as desired.

iv) ROM Vs. PROM

Answer:-

ROM: It is read only semiconductor memory. It is designed for the reading only. It is non-volatile; means it does not lose their contents on power failure. It is written only once at the time of manufacture. It is used by system program only.

PROM: It stands for programmable ROM. It is also non-volatile. It is written only once by special writing device.

- It is possible for a user to customize a system by storing own program in a PROM chip.
- Once a program has been written on to a PROM chip, the recorded information cannot

once by special writing device.

- It is possible for a user to customize a system by storing own program in a PROM chip.
- Once a program has been written on to a PROM chip, the recorded information cannot be changed i.e., the PROM becomes a ROM and it is only possible to read the stored information.
- PROM is also a non-volatile memory. It stored information remains even if power is switched off.

(c) Convert the following number as stated

(i) Decimal 64.006125 to binary

Answer: To convert a decimal number to binary, separate the integer part and the fractional part and convert each part separately.

Integer Part (64):

Divide the integer by 2 and record the remainder

Continue dividing the quotient by 2 until the quotient is 0.

The binary representation is the remainders read from bottom to top.

$$64 \div 2 = 32 \text{ remainder } 0$$

$$32 \div 2 = 16 \text{ remainder } 0$$

$$16 \div 2 = 8 \text{ remainder } 0$$

$$8 \div 2 = 4 \text{ remainder } 0$$

$$\begin{aligned}
 4 \div 2 &= 2 \text{ remainder } 0 \\
 2 \div 2 &= 1 \text{ remainder } 0 \\
 1 \div 2 &= 0 \text{ remainder } 1 \\
 \text{Fractional Part } (0.005125):
 \end{aligned}$$

Multiply the fractional part by 2.
The integer part of the result is the next binary digit.

Continue with the fractional part of the result until you reach the desired precision.

$$\begin{aligned}
 0.005125 \times 2 &= 0.01025(0) \\
 0.01025 \times 2 &= 0.0205(0) \\
 0.0205 \times 2 &= 0.041(0) \\
 0.041 \times 2 &= 0.082(0) \\
 0.082 \times 2 &= 0.164(0) \\
 0.164 \times 2 &= 0.328(0) \\
 0.328 \times 2 &= 0.656(0) \\
 0.656 \times 2 &= 1.312(1) \\
 0.312 \times 2 &= 0.624(0) \\
 0.624 \times 2 &= 1.248(1)
 \end{aligned}$$

So, the binary representation of the fractional part is approximately 000000010100.

Combining both parts:

$$64.005125 \approx 1000000.000000010100 \quad (\text{binary})$$

(ii) Decimal 2376 to hexadecimal

Answer: To convert a decimal number to hexa-

decimal, divide the number by 16 and record the remainder.

$$2376 \div 16 = 148 \text{ remainder } 8$$

$$148 \div 16 = 9 \text{ remainder } 4$$

$$9 \div 16 = 0 \text{ remainder } 9$$

So, the hexadecimal representation is 948.

(iii) Character A and Z to ASCII and Unicode
Hexadecimal CFE9A to binary.

Answer: Character A:

ASCII: 65

Unicode Hex: U+0041

Character Z:

ASCII: 90

Unicode Hex: U+005A

To convert a hexadecimal number to binary,
Convert each hexadecimal digit to its 4-bit
binary equivalent:

$$C = 1100$$

$$F = 1111$$

$$E = 1110$$

$$9 = 1001$$

$$A = 1010$$

Combining these:

$$CFE9A \text{ (hex)} = 1100 \ 1111 \ 1110 \ 1001 \ 1010 \text{ (binary)}$$

d) What is an instruction? What are its components? What is the role of an instruction in a computer? Explain with the help of an example. Where does the instruction reside at the time of execution.

Answer: An instruction specifies the operation performed by CPU. An instruction tells the processor what task is to be performed and what micro operations need to be completed to perform the task.

Every instruction is comprised of two parts: opcode and operands.

- The opcode specifies the operation to be performed.
- Operands provide the data on which the operation is to be performed.

Example

operation	Opcode
Addition	000
Subtraction	001
Multiplication	010
Division	011
Modulus	100
Complement	101
Bitwise AND	110
Bitwise OR	111

Instructions are stored in CPU register during operation.

- e) A 2.5 inch diameter disk has 8 platters with each platter having two data recording surfaces, each platter on disk has 4048 tracks, each track has 400 sectors and one sector can store 1MB of data. Calculate the storage capacity of this disk in Bytes. If this disk has a seek time of 2 milli-seconds and rotates at the speed of 6000 rpm, find the Access time for the disk. Make suitable assumptions, if any.

Answer:-

$$\text{Surface } m = 8 \times 2 = 16$$

$$\text{Track } (t) = 4048$$

$$\text{Sector per track } (p) = 400$$

$$\text{Byte per sector } (s) = 1\text{MB} \times 1024 \times 1024 \text{ bytes}$$

$$\text{Capacity} = mtps$$

$$= 16 \times 4048 \times 400 \times 1\text{MB} \times 1024 \times 1024 \text{ bytes}$$

$$\text{Access time} = \text{seek time} + \text{latency time}$$

$$\text{Latency time} = \left(\frac{1}{(\text{RPM}/60)} \right)^{0.5}$$

$$= (1/6000/60)^{0.5}$$

$$= (1/100)^{0.5} \times 1000 \text{ ms}$$

$$= 5 \text{ ms}$$

$$\text{Access time} = 2\text{ms} + 5\text{ms} = 7\text{ms}$$

(f) What are the uses of various components of motherboard of a computer? List at least four output devices and ports to which these devices can be connected. Explain the characteristics of these output devices and ports.

Answer: Motherboard holds some of the most important components of the computer system. It acts as a base for these components. It is also known as system board, main board etc. In a typical computer microprocessor, main memory and other components are connected to the motherboard. Motherboard also provides connectors for several other devices.

• ATX Power Connector: Advanced Technology Extended (ATX) power connector is used to connect computer's power supply to motherboard.

• AGP Slot: Accelerated Graphics Ports (AGP) is a point-to-point channel which is used to attach a video card to a motherboard.

• CD-In header: At this CD drive is plugged in or connected.

• HDD Headers: Hard Disk Drive header is used for connecting to hard disk.

• PCI Slots: It is used for connecting the PCI

card.

- USB Headers:- It is group of pins to which an internal USB cable can be attached to provide extra USB ports.

(g) What are the uses of following softwares:

(i) Data Compression Utility

Answer:- Data compression is the process in which information is encoded with lesser bits in compare to the original representation.

ii) Media Player

Answer:- It is one of the other common example of application software. It is an application which is used to play multimedia files.

iii) Disk Defragmenter

Answer:- It is used to rearrange the files and unused space on the hard disks that program run faster.

iv) Disk checker

Answer:- Disk checkers are used to check the integrity of the hard disk and pen drive/flash drive. CHKDSK is a command which is used for this purpose. It fixes the logical file system errors found in the disk/drive.

Question:

(a) Why do you need virus detection software? What are their drawbacks? What are the techniques to identify a virus? List any 8 latest virus for desktop systems.

Answers: Computer viruses are software programs that are deliberately designed to interfere with computer operation, record, corrupt or delete data, or spread themselves to other computers and throughout the Internet. Virus scanning software is utility programs designed to protect your computer from computer viruses, worms and Trojan horses.

- Antivirus software can impair a computer's performance. Active anti-virus programs can cause conflicts with other programs.
- Most popular anti-virus programs are not very effective against new viruses.
- Slow down the system.

Techniques:

- Signature based detection: This is the most common method. It compares the contents of the infected file to a known pattern of data. Because viruses can embed themselves in existing files, the entire file is searched.
- Heuristic-based detection: This method is primarily used to identify unknown viruses by looking for malicious code or variations

of such code.

- File emulation:- This is another heuristic approach in which the infected program is run in a virtual environment and the actions it performs are recorded.

Backup

Backup refers to making copies of data so that these additional copies may be used to restore the original after a data loss event. All types of data could be backed up like pictures, word documents, files, executables or an entire database.

- (b) Consider that you have to run several computer programs simultaneously on a computer. Each program takes input from a file and output information on a printer. How do different components of an Operating System (like memory management, I/O management, process management, file system and user interface) will help in execution of these programs.

Answer:- Process Management:-

A process is a program under the execution. It is the job, which is currently being executed by processor (CPU). OS manages process of jobs.

Memory Management:-

OS take care of memory

allocation and de-allocation of main memory to various processes. It allocates the memory space in main memory and secondary memory to store the system program, user program and data.

Input Output Management

OS co-ordinates the various input and output devices such as terminals, printers, disk drives, tape drives. It controls I/O Devices.

File Management :-

As we know, data are stored in a file and files are stored in a device. All operation related to file are managed by operating system such as data storage, data retrieval, updation, deletion and other activities.

Scheduling Management

Operating system establishes process priority for the task execution.

Security Management

Operating system provides data security and data integrity. That is, it protects data and programs stored in computer from unauthorized access.

c) Explain the differences between procedural programming and object oriented programming with the help of one example program of each.

Answer: Procedural programming is a programming paradigm that follows a linear top-down approach where the program is divided into procedures or functions. These procedures perform operations on data and the data is passed between these functions to achieve the desired result.

Example Program:- Calculate the area and perimeter of a rectangle.

```
#include <stdio.h>
// function to calculate the area of a rectangle
float calculateArea(float length, float width) {
    return length * width;
}
// function to calculate the perimeter of a rectangle
float calculatePerimeter(float length, float width) {
    return 2 * (length + width);
}

int main() {
    float length, width;

    // Input length and width
    printf("Enter the length of the rectangle:");
    scanf("%f", &length);
```



```
printf("Enter the width of the rectangle:");  
scanf("%f", & width);
```

```
// Calculate and display the area and perimeter  
float area = calculateArea(length, width);  
float perimeter = calculatePerimeter(length,  
width);
```

```
printf("Area: %.2f\n", area);  
printf("Perimeter: %.2f\n", perimeter);  
return 0;
```

}

Object Oriented language

Object Oriented Programming is more abstract than procedural programming because it looks for patterns and reusability. The same code can be loaded and executed many times to accomplish a task without having to retype it.

Before we consider above example in object oriented, let us understand few terms and concepts associated with object oriented programming. There are three main concepts that any language needs to support to be an object oriented language.

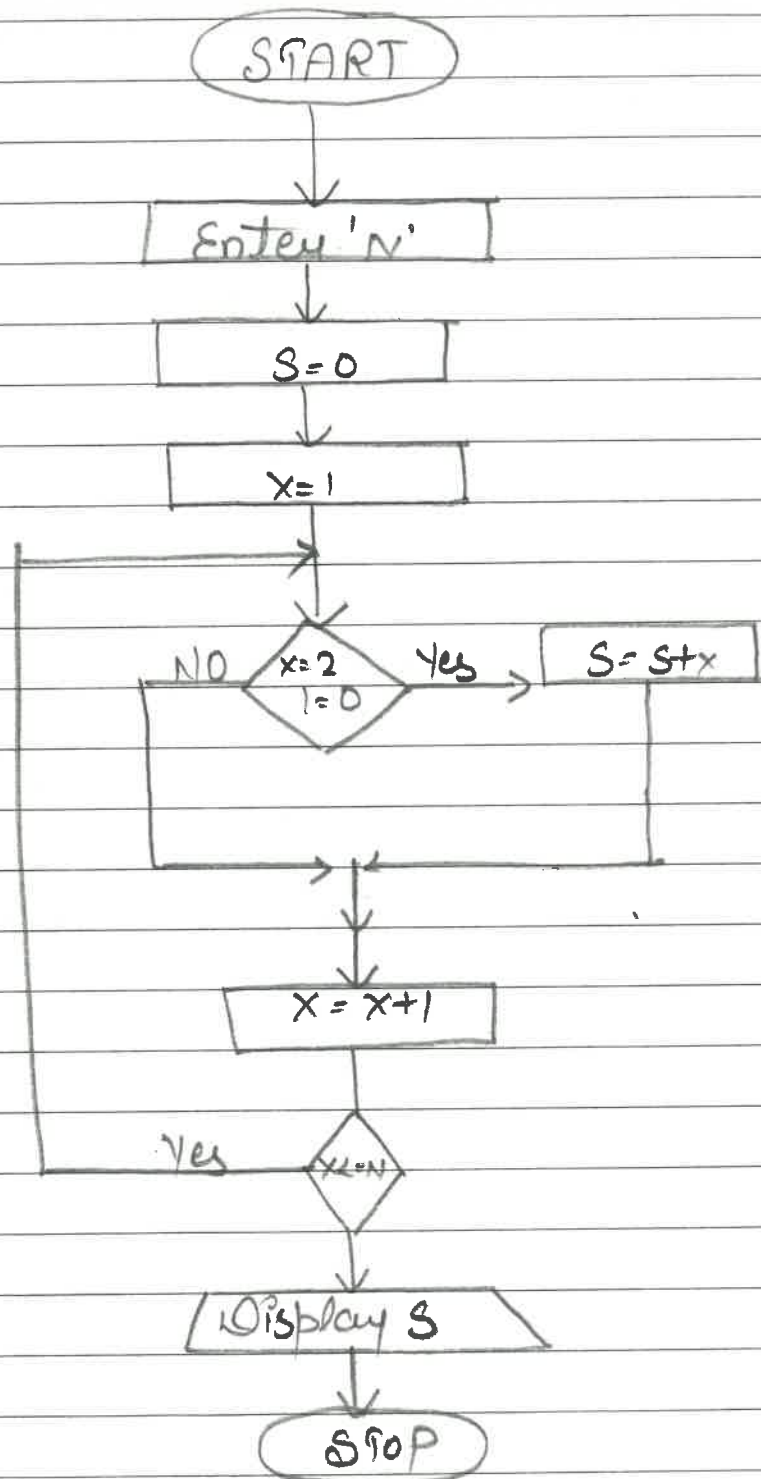
Encapsulation - Is a mechanism through which a protective wrapper is created to hide the implementation details of the object and the only thing that remains externally

visible is the Interface of the object.

Inheritance :- Is the process by which a new class is created using an existing class. It is a way to compartmentalize and reuse code since it allows classes to inherit commonly used state and behavior from other classes.

Polymorphism :- Polymorphism is the characteristic of being able to assign a different meaning specifically, to allow an entity such as a variable, a function, or an object to have more than one form. It is the ability to process objects differently depending on their data types and to redefine methods for derived classes.

- d) Draw a flow chart of a program that adds N even numbers starting from 1. The value of N should be input by the user.

Answer:

- e) List the elements of a programming language. Explain the terms data, type, expression, assignment, and logical, relational and equality operators with the help of an example each.

Answer: Elements of Programming Language

- 1) Variables, Constants, Data type, Array and Expression
- 2) Conditional and Looping Statement
- 3) Subroutine and function

Data Types: Anything that is processed by a computer is called data. There are different types of data that can be given to the computer for processing. A data type is a classification identifying the type of data. It determines the

Possible values for that type

Operations that can be performed on values of that type,

The way values of that type can be stored in memory,

In each programming language there are some primitive data types. For example in the C programming language they are: Please note that these sizes can be compiler or machine dependent in the case of this language. For other languages such as Java, the sizes are defined by the language itself.

int, both signed and unsigned integer, 2 bytes
in size.

float, floating point numbers, up to 4 bytes

In size.

double, floating point number with double precision. These are organised in 8 bytes (64 bits).

Char, Character type size of 1 byte (8 bits)
It is used to form the string i.e. sequence of characters.

Expression

An expression is a combination of variables, constants and operators written according to the syntax of the programming language. In the C programming language every expression evaluates to a value i.e., every expression results in some value of a certain type that can be assigned to a variable. Every computer language specifies how operators are evaluated in a given expression. An expression may contain

- i) arithmetic operator
- ii) relational operator
- iii) logical operator

Assignment

It is composed of variable name, an assignment operator of the language and a value or variable or some expression as per composition allowed based on rules defined in grammar.

e.g $\text{temp} = 5;$
 $\text{temp} = \text{temp} + 1;$

This means to add 1 to the current value of the variable temp and make that the new contents of the variable temp

$\text{temp} = a + b;$

Logical, relational and equality:

These types of expression result in a Boolean representation i.e. each expression will result in either True or false. It is composed of operands and relational/logical/equality operator.

The following is the list of operators in the C programming language.

$=$ (equal to)

$\neq$ (Not equal to)

$<$ (less than)

$\leq$ (less than equal to) $>$ (greater than)

$\geq$ (greater than equal to) $\&\&$ (logical AND)

$\|\|$ (logical OR)

$!$ (logical NOT)

Relational expressions result in one of the truth value either TRUE or FALSE. They are

capable of comparing only two values separated by any valid relational operator.

e.g.

Let $x=1, y=3$

$x=1$ /* evaluates to true as x has value 1 */
 $=y$ /* evaluates to false */

$x < y$ /* evaluates to true */

$(x < 2) \&\& (y > 5)$ /* evaluates to true */

f) What are the phases of project development in which project management software can help. Explain with the help of examples.

Answer: Project management software can assist in all phases of project development, helping to overcome the primary challenge of meeting project goals within constraints like scope, time and resources.

Project Initiation Stage:

During this phase, software helps define project objectives, identify stakeholders and establish a project charter. Tools like Trello or Asana enable collaborative brainstorming and documentation, ensuring all stakeholders are aligned from the start.

Project Planning and Design Stage:-

In this stage detailed project plans, schedules and resource allocations are developed. Software like Microsoft Project or Smartsheet can create Gantt charts, manage dependencies and calculate the critical path, ensuring all tasks are appropriately scheduled and resources are optimally allocated.

Project Implementation Stage:-

Project management software aids in task assignment, progress tracking and resource management.

Project Monitoring and Controlling:-

Software provides dashboards and reporting features to monitor project progress against the plan. Tools like Tableau or Power BI offer real-time insights and performance metrics, allowing for quick adjustments to keep the project on course.

Project Completion Stage:-

In this final phase, software helps compile final reports, manage documentation and gather feedback.

g) Explain the following with the help of an example/diagram, if needed:

i) Development Model for Open Source Software

Answer: The Open Source Development Model is a collaborative approach where multiple developers contribute to a single project. While theoretically, any open source project can have hundreds or thousands of contributors, most are small, involving just a few contributors. Large projects like Linux or Apache may have many contributors but are managed by a few "gatekeepers" or "committees" who control the official source code. This model resembles a free market with specialized oversight rather than a completely open bazaar, ensuring high quality, stable and secure code for the community.

ii) Tools for program development

Answer: Tools for program development are software applications that assist developers in writing, testing, debugging and maintaining code. Common tools include:-

Integrated Development Environments (IDEs):-
Examples are Visual Studio, Eclipse and Pycharm, which offer code editors, debugging tools, and build automation features.

Version Control System (VCS):

Tools like Git and Subversion help manage code changes, enabling collaboration among multiple developers and maintaining a history of change.

Build Automation Tools:

Tools such as Maven, Gradle and Make automate the process of compiling code, running tests and packaging applications.

Debugging Tools:

Tools like GDB and Visual Studio Debugger help identify and fix code errors.

Testing Frameworks:

Tools like JUnit, NUnit and Selenium automate the testing process to ensure code quality.

iii) Use of Functions and Macros

Answer:

Functions and macros streamline data handling and automation in Excel.

• Advanced Functions:

MATCH: Finds the position of a value.

Example: - MATCH("Apple", A1:A10, 0) returns the position of "Apple".

INDEX: - Retrieves a value from a specific position. Example: - INDEX(B1:B10, 3) returns the 3rd value in B1.

SEARCH: - Locates a substring within text. Example: - SEARCH("cat", "catalog") finds "cat" starting at position 1.

REPLACE: - Substitutes part of text. Example: - REPLACE("Hello World", 7, 5, "Universe") changes "World" to "Universe".

Database functions

DGET: - Extracts a record matching criteria. Example: - DGET(A1:D10, "Salary", B1:B2).

DCOUNT: - Counts numeric entries. Example: - DCOUNT(A1:D10, "Salary", B1:B2).

DSUM: - Sums value based on criteria. Example: - DSUM(A1:D10, "Sales", B1:B2).

DPRODUCT: - Multiplies values based on criteria. Example: - DPRODUCT(A1:D10, "Quantity", B1:B2).

Macros :- Automate repetitive tasks using VBA. Example VBA code to format cells:

Sub FormatCells()

With Range ("A1:A10")

• Font.Bold = True

• Interior.Color = RGB(255, 255, 0)

'Yellow background

End With

End Sub

Qv) Database and Database Management System

Answer :- Database :- A structured collection of data stored electronically. It organizes data in tables, rows and columns, for example, a customer database might include tables for customers, order and products with each table containing related data.

Example :- Stores customer information, order history and inventory.

Database Management Systems (DBMS) :-

Software that manages and interacts with databases, allowing users to create, read, update and delete data. It ensures data integrity, security and efficient querying. Examples include MySQL, Oracle and Microsoft SQL Server.

Example :- manages data access, ensure accurate data retrieval and supports complex queries like retrieving all orders from a specific customer.

Question

(a) Explain the characteristics of any two guided and any two unguided channels for data transmission.

Answer:- Guided Channels:-

1) Twisted Pair Cable:-

Physical Structure:- Consists of pairs of insulated copper wires twisted together.

Transmission Speed and Distance:- Suitable for short to medium distances with moderate data rates.

Cost:- Relatively inexpensive compared to other guided media.

2) Optical Fiber Cable:-

Physical Structure:- Uses glass or plastic fibers to transmit data as light signals.

Transmission Speed and Distance:- Supports high speed data transmission over long distances with minimal signal loss.

Cost:- More expensive than twisted pair cables but offers superior performance.

Unguided Channels :-

1) Radio Waves :-

Transmission

Range: Suitable for both short range and long range communication.

Susceptible to interference from other radio frequency devices and environmental factors.

2) Microwaves :-

Use high frequency electromagnetic waves for point-to-point communication.

Suitable for long distance communication with high data rates but generally require repeaters for extended ranges.

- (b) Four branch offices of an organisation are located in four major cities of a vast country. Explain the characteristics of the network that will be needed for every branch office. All the four branches offices network should also be connected by another network. Explain the characteristics of this network also.

Answer :- To connect four branch offices located in different cities of a vast country, two types of networks are needed: a local Area Network (LAN) at each branch and a Wide Area Network (WAN) to connect

all branches.

1) Local Area Network (LAN)

- It connects computers in a single building, block or campus, i.e. they work in a restricted geographical area.
- LANs are private networks, not subject to tariffs or other regulatory controls.
- LANs operate at relatively high speed when compared to the typical WAN.

2) WAN

- It generally covers large distances.
- Communication medium used are satellite, public telephone networks which are connected by routers.
- Routers forward packets from one to another on a route from the sender to the receiver.

c) What is Internet & what are the major protocols used on Internet & What is an IP address? How can an IP address be related to a web address? Explain with the help of an example.

Answer The Internet is a global network of interconnected computers and servers that communicate using standardized protocols to share data and resources. It enables access to websites, emails and various online

services.

Major Protocols Used on the Internet:-

- HTTP/HTTPS :- Used for transferring web pages from servers to browsers.
- FTP (File Transfer Protocol) :- Used for transferring files between computers.
- SMTP (Simple Mail Transfer Protocol) :- Used for sending emails.
- POP3/IMAP (Post Office Protocol / Internet Message Access Protocol) :- Used for retrieving emails from servers.
- TCP/IP (Transmission Control Protocol / Internet Protocol) :- The fundamental suite of protocols for Internet communication, ensuring reliable data transmission.

IP Address:- An IP Address is a unique numerical identifier assigned to each device connected to a network.

Relation to a Web Address:- A web address maps to an IP address using the Domain Name System (DNS). For example, the URL `www.example.com` is translated into an IP address like `192.0.2.1` by DNS servers.

Example:-

- URL: `www.google.com`

• IP Address: 142.250.140.78

The IP address directs your request to Google's servers, which then send the requested webpage to your browser.

- d) What are the different features of a browser? If you want to perform online Banking Transactions, what precautions will you take before performing a transaction?

Answer:

Features of a Browser:-

- 1) User Interface: Provides navigation tools like address bars, back and forward buttons, bookmarks and tabs for managing multiple pages.
- 2) Rendering Engine: Converts HTML, CSS and JavaScript into a visual web page display.
- 3) Security features: Includes options for private browsing, managing cookies and warning about potential security threats.

Precautions for Online Banking Transactions:

- 1) Use a secure connection: Ensure you're using HTTPS, indicated by a padlock icon in the address bar, to encrypt data between your browser and the bank's server.
- 2) Update software: Keep your browser

and operating system up to date to protect against vulnerabilities.

3) Use strong Passwords :- Create complex and unique passwords for your banking account and avoid using the same password for multiple sites.

e) Describe the process of creating a web email account. What are the different components of a mail message? Explain with the help of a diagram.

Answer:- Start the browser and Open the Gmail Homepage www.gmail.com. The screen similar to figure 10 may appear on your system. Select the "Create an Account" button on the right.



Figure 10: The Screen for creating an account

Then following page will open

Create an Account

After Google searches, gives you access to their site. Google account. It gives access to many services. You can sign up for it.

Get started with Gmail

First Name:

Last Name:

Gender & Birth Date:

Male Female (Select your gender)

Birth Date:

Month Year

Choose a password:

Re-enter password:

Strong password

Weak password

Security Question:

Answer:

Secondary email:

Location:

What's your phone:

What was your first phone number?

If you forget your password, we will ask you the answer to your security question. Learn more

Phone number

Email address (optional)

This address is used to notify you about your account. You can use another email address. You may have trouble with it. Learn more

Phone

Type the characters you see in the picture below

opical

Sign up

or sign in with Google

Terms of Service

Please check the Google Account information you've entered above and then to change anything you like, and review the Terms of Service below.

With Gmail, you won't see blinking banner ads. Instead, we display ads you might find useful that are relevant to the content of your messages. Learn more

Terms of Service:-

Please check the Google Account Information you've entered above, and review the Terms of Service below.

With Gmail, you won't see blinking banner ads. Instead, we display ads you might find useful that are relevant to the content of your messages. Learn more

I accept. Create my account.

Figure 11: The screen for entering information and creating an account

Fill all the fields and click "I accept. Create my

account."

If you have not made any mistake then your account will be created, otherwise the error will be displayed. Correct the errors and try again. Now you have created your account.

Now you can access your account. An email account, in general has the following folders:

Inbox:- Inbox is the main folder in your email account. It contains all the email that have arrived in your email account. You can click on inbox to see the mails that you have not read as well as the mails that you have already read.

Sent Mail:- It shows all the emails sent by you from/ from your email account.

Drafts:- This folder stores those messages that you have created but have not been sent by you so far. These messages are saved by you for more work.

Spam:- Spam is unsolicited emails or junk mails. It is generally e-mail advertising sent to group of people. We can also term spam as unwanted emails. Spam mails are also a big cause of computer viruses.

Trash :- Any deleted mail is put in the trash folder. Trash folder allows you to get back an email which have already been deleted. But it is important to know that you can get back the mails only within few days from trash after its deletion. After few days, mails are permanently deleted from trash folder.

An email account, in general has the following options:

Compose Mail :- Composing is addressing, writing and sending an email message. By clicking on the Compose Mail button a window appears where we can write our message in the message box and the email addresses of the person we want to send the mail.

Contacts :- The contacts helps you to find email address of a person whom you have saved in your contact list. We can also quickly find email conversations associated with a contact and store additional information about our other persons whose email id is stored in our contact.

Now, if you want to read your mails then click "Inbox" figure 13

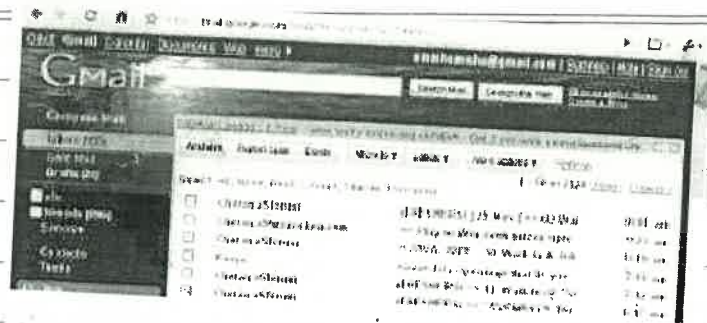


Figure 13: Inbox

If you want to send email then click "compose mail" and screen as shown in figure 14 will appear.

In the To field you have to put the address of the receiver. In case you want to send email to more than one receiver then put commas between their email addresses or you can use "add cc" or "add bcc".

Cc stands for Carbon copy the persons whose address is listed in this field will receive carbon copy of the message and Bcc means blind carbon copy similar to Cc but only difference is that the recipient who had got Bcc is invisible to other recipients.

Subject: The main heading of your mail i.e. It will explain that the mail is regarding which issue. Please note that the subject should be descriptive of the mail to help the receiver understand what mail is about without having to open the mail.

Text Area: The message is written in this area.

When all required fields are filled as shown in figure 14, you can send the mail by clicking the send button.

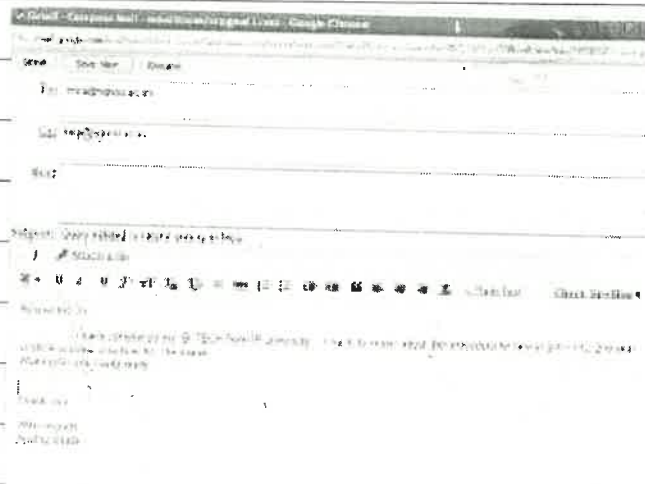


Figure 14: Compose mail window

Besides sending or receiving mails, Emails services have other features. Contacts is one such list as explain earlier. Contact can be organized in different groups like office, school, relatives etc.

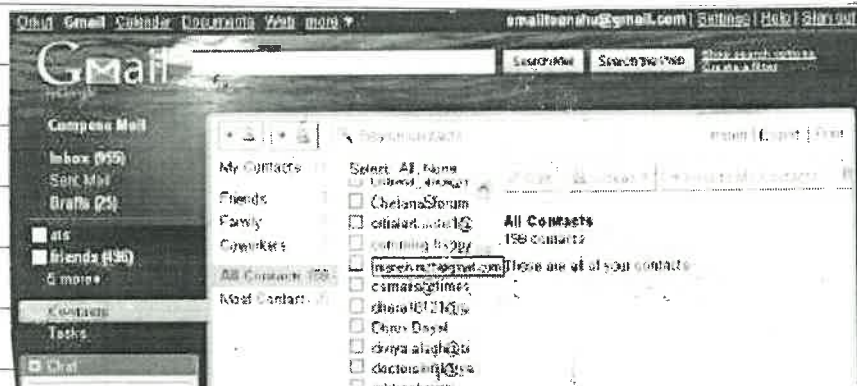


Figure 15: Contacts

f) Explain the following in the context of Internet and its applications, giving their features and uses:

(i) Online Education

Answer: Features:

- Accessibility: Offers learning materials and resources from anywhere with an internet connection, breaking geographical barriers.
- Flexible Scheduling: Allows learning to access courses and complete assignments at their own pace, accommodating various schedules.
- Diverse Learning Resources: Includes video lectures, interactive quizzes, discussion forums and downloadable materials.
- Interactive Tools: Utilizes tools such as virtual classrooms, live webinars and collaborative platforms to facilitate real time interactions and feedback.
- Scalability: Can accommodate a large number of students simultaneously, providing a cost-effective solution for educational institutions.
- Personalized Learning: Provides adaptive

learning systems and personalized feedback to cater to individual learning style and needs.

Uses :-

- Academic Courses :- Universities and colleges offer degree programs, diplomas and certifications online.
- ~~Per~~ Professional Development :- Platforms like Coursera and LinkedIn Learning offer courses for skill enhancement and career advancement.
- Skill Training :- Websites like Udemy provide courses on specific skills such as programming, graphic design, or languages.

(ii) e Commerce

Answer :-

FEATURES :-

- Online Transactions :- Enables buying and selling of goods and services over the Internet using secure payment gateways.
- Product Listings :- Provides detailed product descriptions, images and customer reviews to aid purchase decisions.

- Shopping Cart :- Allows users to select and store items for purchase before checkout.
- Order Tracking :- Provides real time tracking of shipments and order status.
- Personalized Recommendations :- Uses algorithms to suggest products based on user behavior and preferences.

Uses :-

- Retail Shopping :- Websites like Amazon and Walmart allow customers to purchase a wide range of products online.
- Digital Services :- Platforms such as Netflix and Spotify offer subscription-based digital media services.
- B2B Transactions :- Businesses use platforms like Alibaba for bulk purchasing and procurement.
- Marketplaces :- Websites like eBay and Etsy provide a platform for individuals and small businesses to sell products.